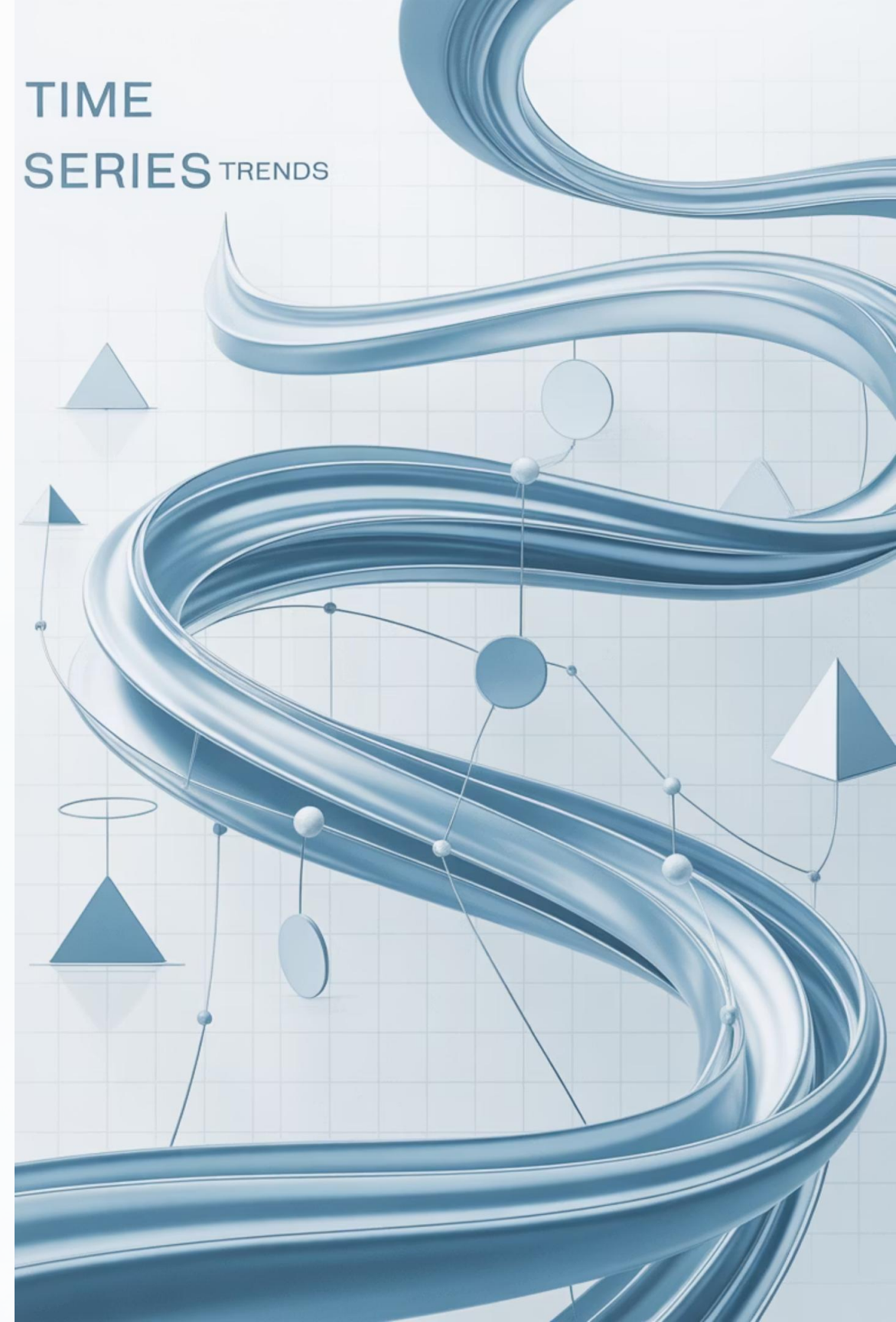


Time Series Fundamentals

TIME
SERIES TRENDS



Time Series Fundamentals



Foundations

Grasp the fundamental concepts and principles underlying time series data.



Tools

Familiarize yourself with essential statistical and computational tools.



Models

Understand and apply various time series models, including ARIMA.



Forecasting

Learn techniques to predict future values based on historical data.

Growth
Potential

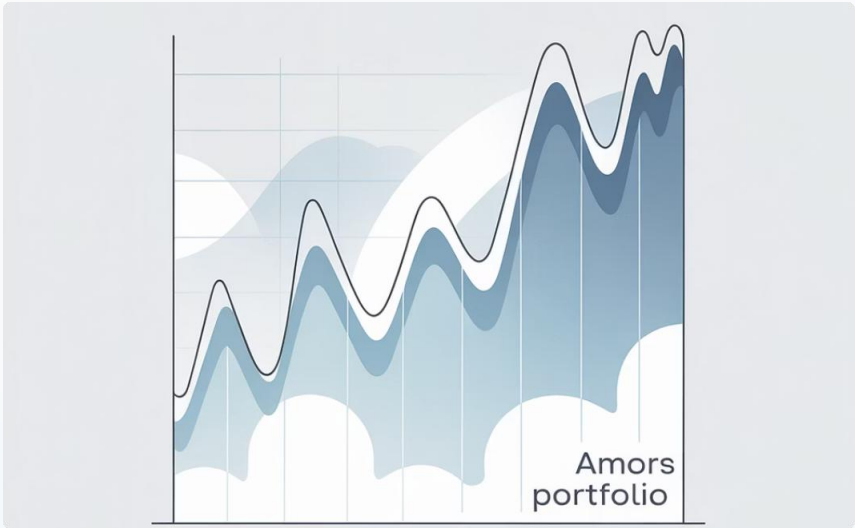


What is a Time Series?

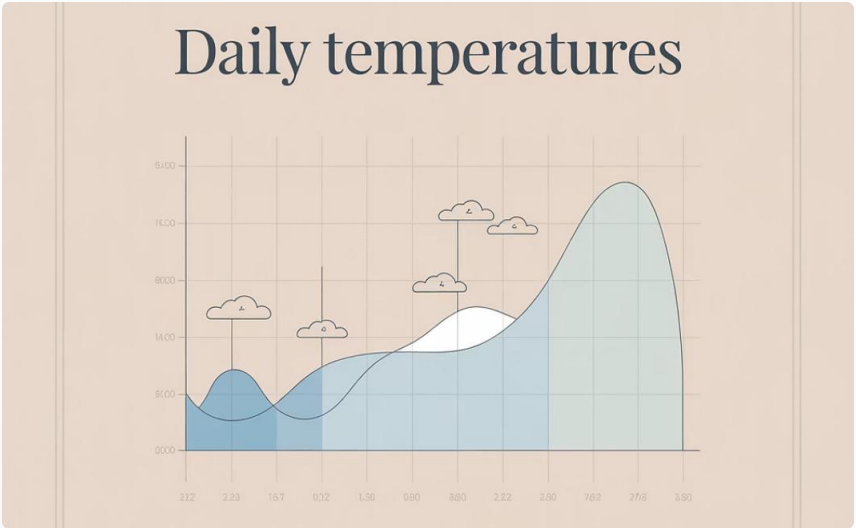
A **time series** is a sequence of data collected over time. The key is that order matters — yesterday's stock price is related to today's.

Unlike cross-sectional data, ignoring time ordering can lead to false conclusions.

Daily fluctuations in market values

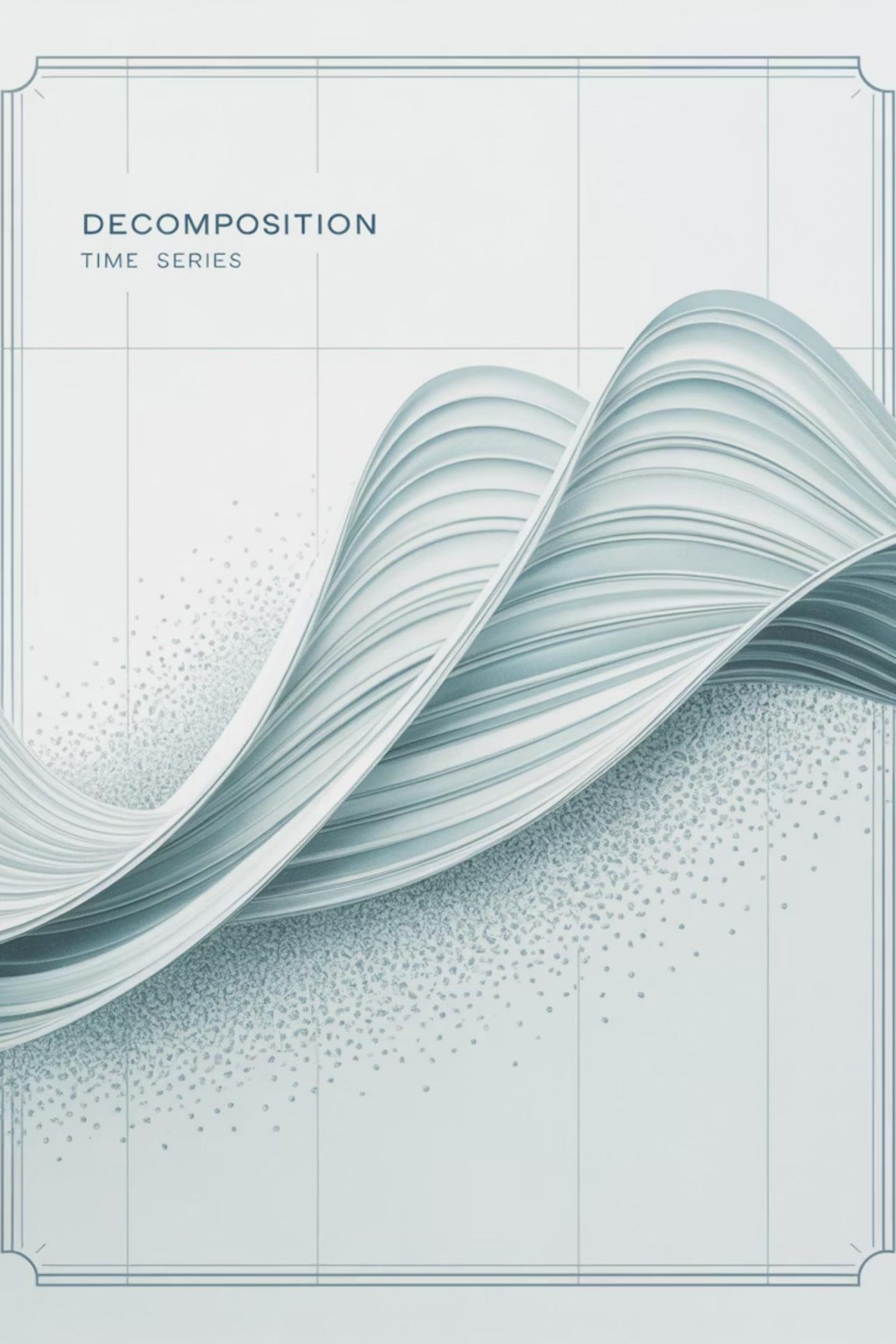


Weather patterns over time



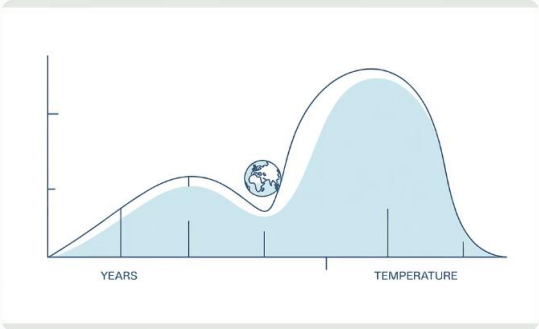
GDP Growth





Components of Time Series

Most time series can be decomposed into four fundamental components, revealing their underlying structure:



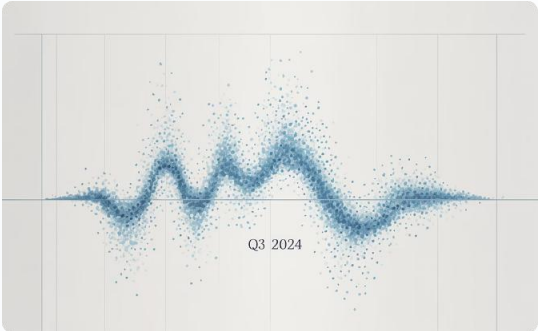
Trend: The long-term direction or movement of the series, like rising global temperatures.



Cyclicity: Fluctuations that are not fixed in period or amplitude, often associated with economic or business cycles.



Seasonality: Regular, predictable patterns that recur over a fixed period, often tied to calendar cycles like holiday sales.



Irregular/Noise: The random, unpredictable component that remains after accounting for trend, seasonality, and cyclicity.

Understanding these components helps us distinguish between predictable patterns and random variations, crucial for effective analysis and forecasting.

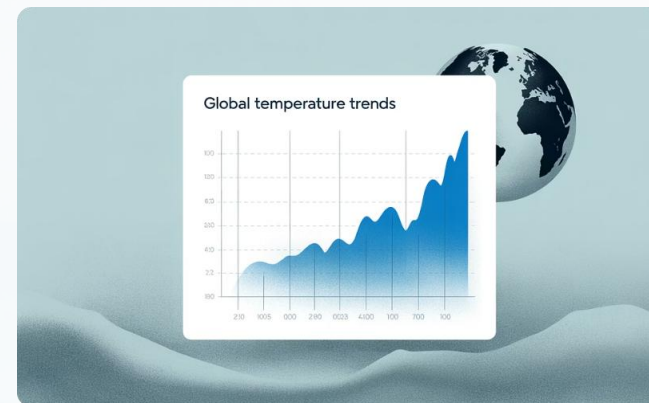
Applications of Time Series

Time series analysis is central to many disciplines. The unifying principle is: *past values contain information about future ones.*



Forecasting Demand

Predicting electricity usage, retail sales, and resource needs



Climate

Modeling temperature changes and weather patterns



Finance

Managing investment risk and predicting market movements



Health

Monitoring epidemics and patient vital signs

Stationarity

A series is **stationary** if its mean, variance, and autocovariance don't change over time.

Types of Stationarity

- **Strict stationarity:** Full probability distribution is invariant to time shifts
- **Weak stationarity:** Mean and variance remain stable over time

Importance

Most time series models assume stationarity. Non-stationary series (like stock prices) must be differenced or transformed before modeling.



Autocorrelation & Partial Autocorrelation

Autocorrelation Function (ACF)

Shows correlation between a series and its lags

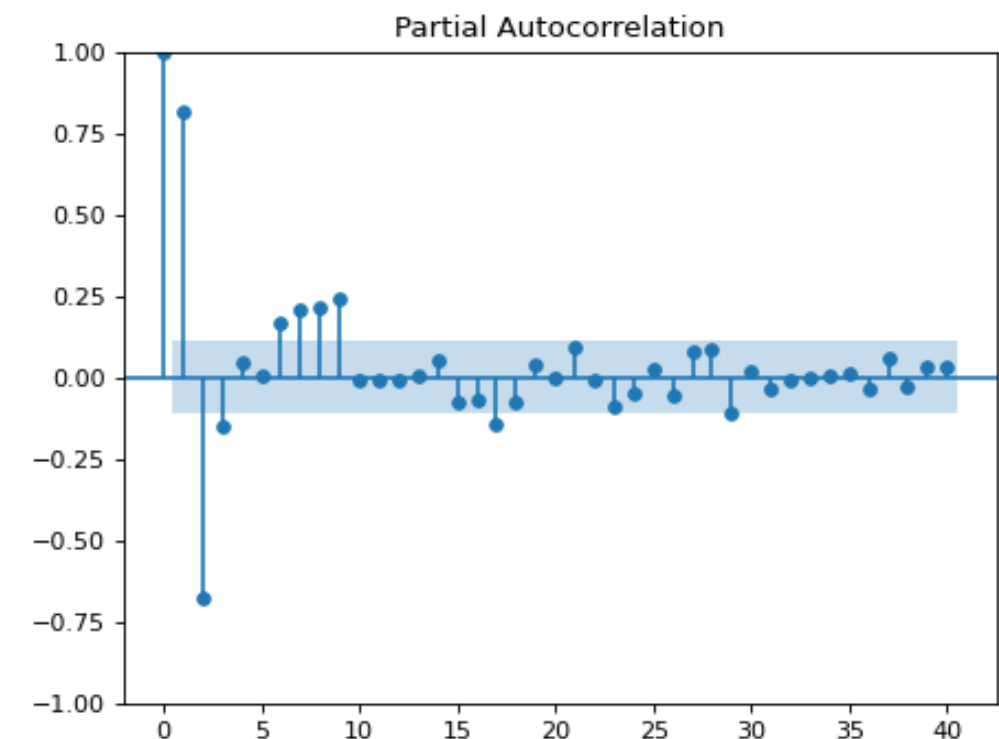
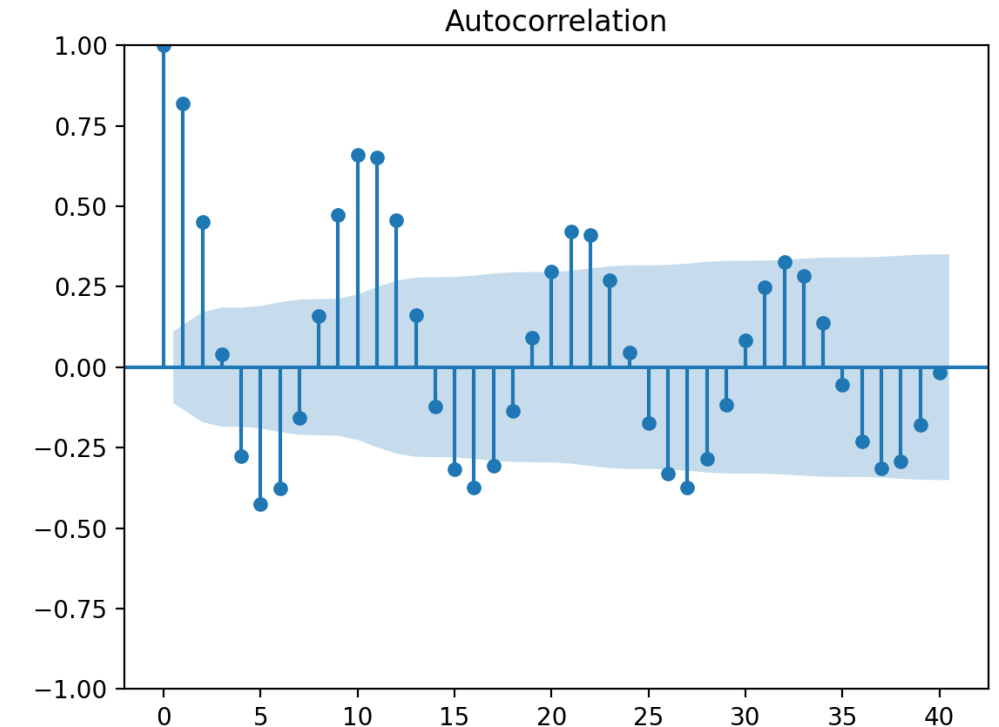
- Measures total dependence
- ACF cutoff suggests MA model

These statistical tools are essential for identifying dependence structure and guiding ARMA/ARIMA order selection, as visually represented by their plots.

Partial Autocorrelation Function (PACF)

Shows correlation at specific lag controlling for shorter lags

- Measures direct dependence
- PACF cutoff suggests AR model



White Noise vs Random Walk

White Noise

Pure randomness with:

- Mean zero
- Constant variance
- No autocorrelation

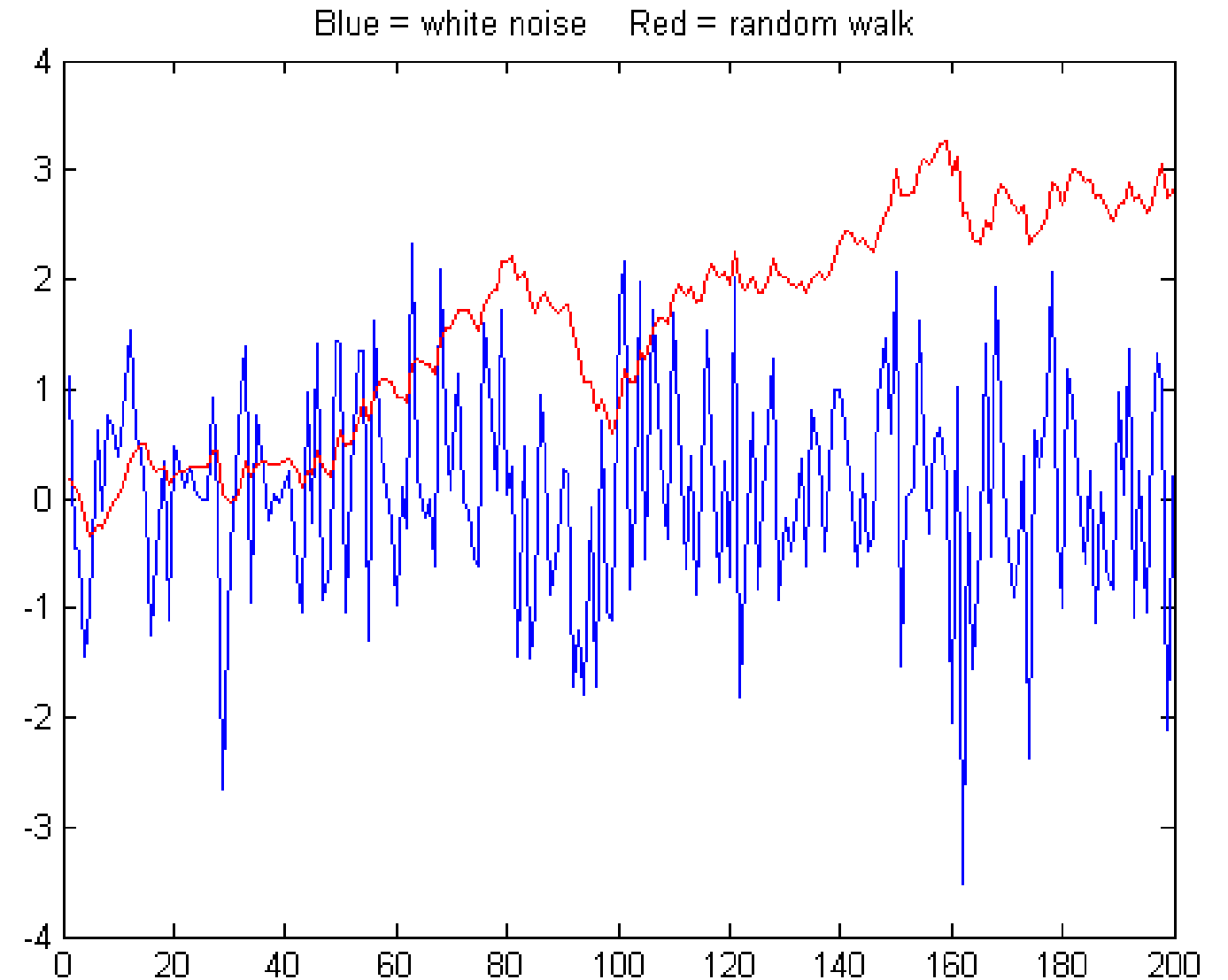
If residuals look like white noise, our model is good.

Random Walk

Today = yesterday + noise

- Non-stationary process
- Variance grows over time
- Stock prices often resemble random walks

These two processes serve as benchmarks for predictability in time series analysis.



Autoregressive (AR) Models

AR(1) Model

$$X_t = \phi X_{t-1} + \varepsilon_t$$

AR models express today as a function of past values plus noise:

- AR(1): current depends on last value
- AR(p): depends on p past values

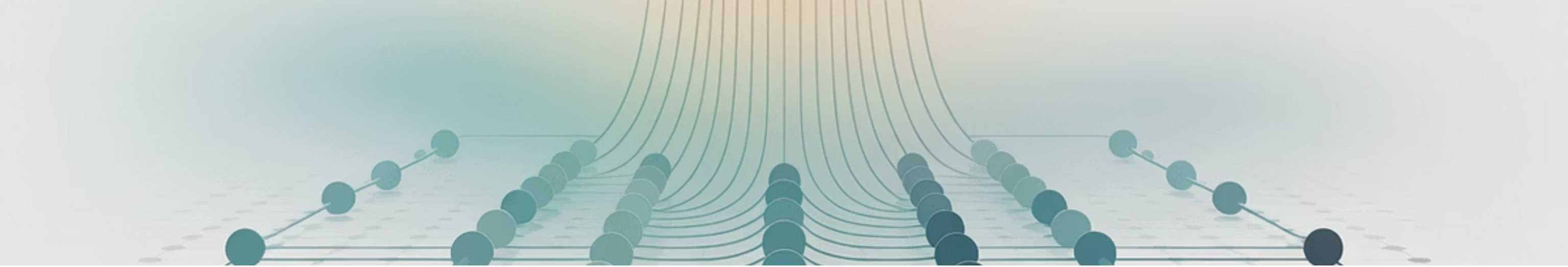
AR models are useful for short-memory processes like inflation rates.

Characteristics

- Captures persistence
- Shows mean reversion
- PACF cuts off at lag p



Autoregressive
Time Series



Moving Average (MA) Models

MA(1) Model

$$X_t = \varepsilon_t + \theta\varepsilon_{t-1}$$

MA models describe dependence on past shocks:

- Always stationary
- Shocks persist for q periods
- ACF cuts off after lag q

Applications

Useful when short-term shocks drive the data, such as:

- Forecasting residual fluctuations
- Modeling temporary effects
- Capturing short-term dependencies

ARMA & ARIMA Models

ARMA(p,q)

Combines Autoregressive and Moving Average components

$$X_t = c + \sum_{i=1}^p \phi_i X_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$$

ARIMA(p,d,q)

Extends ARMA with differencing to remove trends

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$$
$$Y_t = X_t - X_{t-d}$$

This family is widely used in economics, retail, and weather forecasting.

Limitations

- Linear relationships only
- Usually univariate (single variable)
- Assumes constant parameters over time

Seasonal ARIMA (SARIMA)

SARIMA extends ARIMA with seasonal terms:

$$\phi_p(B)\Phi_P(B^s)(1 - B)^d(1 - B^s)^D X_t = \theta_q(B)\Theta_Q(B^s)\varepsilon_t$$

where B is the backshift operator, s = season length (e.g., 12 for monthly data), and P,D,Q represent the seasonal orders.

Key Features

- Captures recurring seasonal dependencies
- Handles both trend and seasonality
- More complex but more realistic for seasonal data

Applications

- Retail sales with annual peaks
- Seasonal climate effects
- Energy consumption patterns
- Tourism demand forecasting

Model Diagnostics

Diagnostics test model adequacy. Residuals should be white noise.

Residual Analysis

Check residual plots and ACF for remaining patterns

Residual Auto-correlation Test

Statistical test for remaining autocorrelation

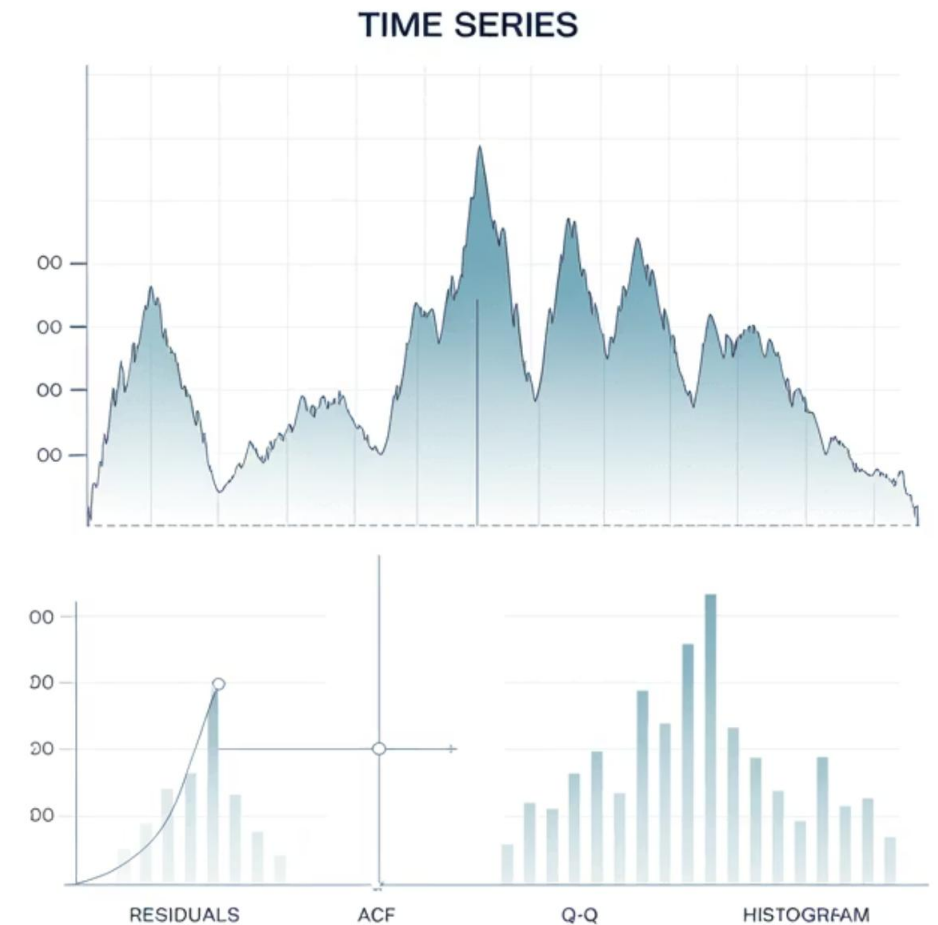
Variance Checks

Test for heteroscedasticity (changing variance)

Linearity Checks

RESET Test

If residuals aren't white noise, the model needs refinement.



Forecasting

Forecast Types

- **Point forecasts:** Best guess of future values
- **Prediction intervals:** Range showing forecast uncertainty

Time Horizons

- **Short-term:** More reliable
- **Long-term:** Accumulate error

Challenges

- Error accumulation over time
- Structural changes break models
- Crises and regime shifts
- Unexpected external factors

Time
Series
Forecast



Exponential Smoothing Methods

Exponential smoothing gives more weight to recent data.

1

Simple Exponential Smoothing (SES)

For data with no trend or seasonality

2

Holt's Method

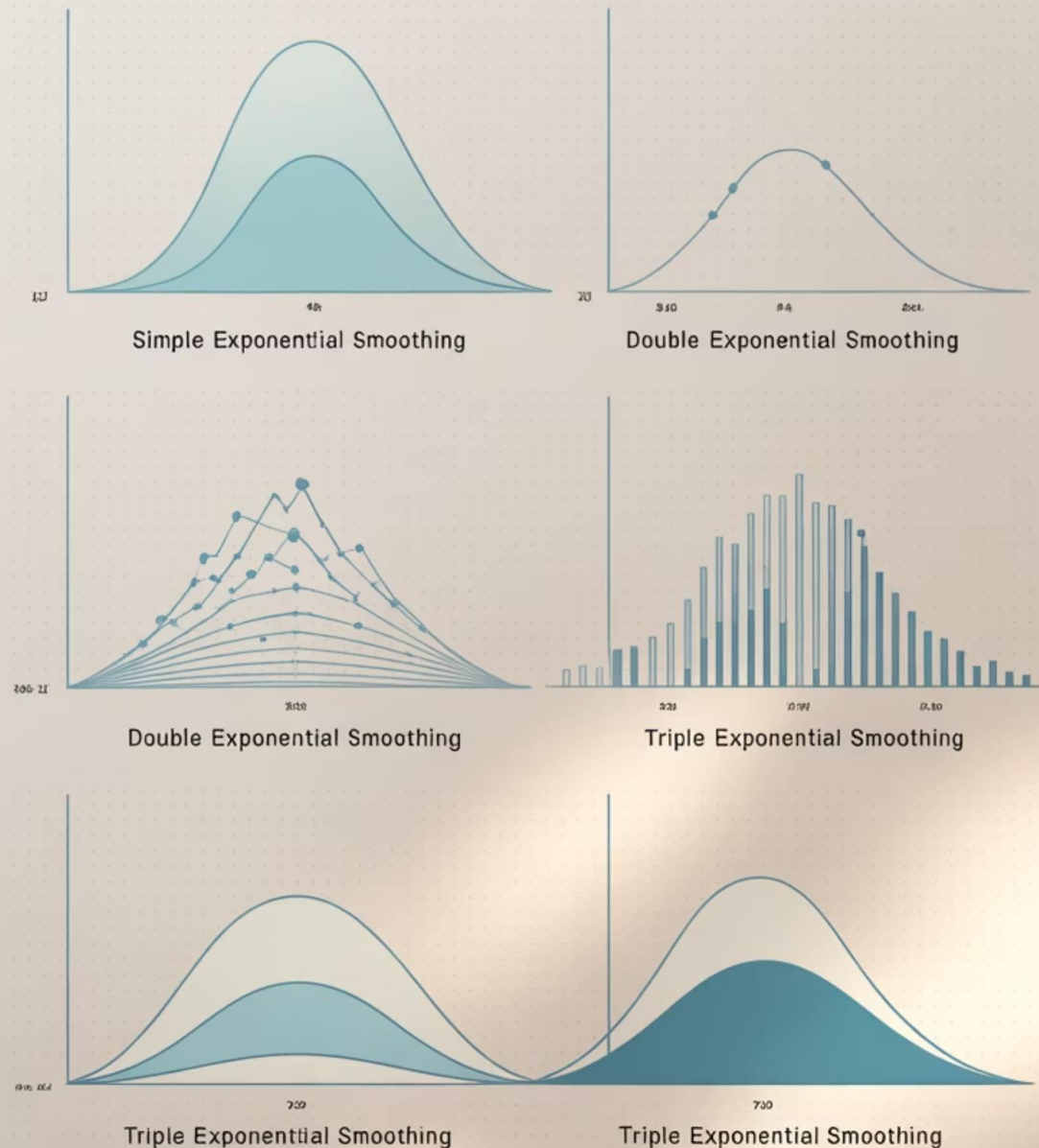
Adds trend component to SES

3

Holt–Winters Method

Adds seasonality to Holt's method

These methods are simple, fast, and often accurate — sometimes rivaling ARIMA in forecasting performance.



Modern Approaches

State-Space Models

- Unifying framework for time series
- Estimated with Kalman filter
- Handles missing data and multiple series

Example: Tracking aircraft position and velocity.

Machine Learning

- Neural networks (LSTMs, RNNs)
- Captures nonlinear patterns
- Handles high-dimensional data
- Sacrifices interpretability for accuracy

Example: Predicting complex sales patterns.

Both approaches extend classical time series analysis, offering new capabilities while presenting different trade-offs.

Common Pitfalls

Frequent mistakes in time series analysis include:



Overfitting

Using too many parameters, capturing noise instead of signal



Ignoring Non-Stationarity

Applying standard methods to non-stationary data without transformation



Spurious Correlation

Mistaking two trending series as causally related



Poor Validation

Failing to test model performance on out-of-sample data

Good modeling requires skepticism and rigor.





Post-Model Testing & Features

After fitting a model, test its robustness:

Residual Tests

- **Stationarity:** Residuals must be stationary
- **Autocorrelation:** No pattern should remain

Distribution Tests

- **Heteroscedasticity:** ARCH LM test for constant variance
- **Linearity**

Residuals should behave like white noise. If not, revise the model.

Summary of Key Takeaways

1 Time Series = Ordered Data

Temporal ordering matters, with observations showing dependence over time

2 Components

Trend, seasonality, cycles, and noise form the building blocks of time series

3 Core Models

AR, MA, ARIMA, SARIMA, and exponential smoothing provide the foundation

4 Diagnostics

Residual analysis is essential for model validation

5 Forecasting

Requires both statistical rigor and judgment